ClipSAT : Multivariable Calculus Partials, Multiple Integrals & Vector Calcu [ANSWER KEY]

Unit 5.01: Triple Integrals, Cylindrical & Spherical Coordinates

Q1. Evaluate iiint_B x y z,dV over the box B=[0,3]times[0,1]times[0,3] .

[ ] A: frac{81}{4}
[ ] B: frac{81}{16}
[x] C: frac{81}{8}
[ ] D: frac{153}{8}

Solution: Iterate: intlimits_{0}^{3}intlimits_{0}^{1}intlimits_{0}^{3} x y z, dx, dy, dz = frac{81}{8} .

Q2. Evaluate iiint_B x + y + z,dV over the box B=[0,3]times[0,2]times[0,1] .

[ ] A: 36
[ ] B: 9
[x] C: 18
[ ] D: 24

Solution: Iterate: intlimits_{0}^{1}intlimits_{0}^{2}intlimits_{0}^{3} leftx + y + zright, dx, dy, dz = 18 .

Q3. Evaluate iiint_B x y z,dV over the box B=[0,2]times[0,1]times[0,2] .

[ ] A: 4
[ ] B: 1
[ ] C: 6
[x] D: 2

Solution: Iterate: intlimits_{0}^{2}intlimits_{0}^{1}intlimits_{0}^{2} x y z, dx, dy, dz = 2 .

Q4. Evaluate iiint_B y z^{2},dV over the box B=[0,1]times[0,1]times[0,3] .

[ ] A: 9
[ ] B: frac{9}{4}
[ ] C: frac{15}{2}
[x] D: frac{9}{2}

Solution: Iterate: intlimits_{0}^{3}intlimits_{0}^{1}intlimits_{0}^{1} y z^{2}, dx, dy, dz = frac{9}{2} .

Q5. Evaluate iiint_B x^{2} z,dV over the box B=[0,1]times[0,1]times[0,3] .

[ ] A: 3
[ ] B: frac{3}{4}
[x] C: frac{3}{2}
[ ] D: frac{9}{2}

Solution: Iterate: intlimits_{0}^{3}intlimits_{0}^{1}intlimits_{0}^{1} x^{2} z, dx, dy, dz = frac{3}{2} .

Q6. Evaluate iiint_B x y + z,dV over the box B=[0,1]times[0,1]times[0,3] .

[x] A: frac{21}{4}
[ ] B: frac{21}{2}
[ ] C: frac{21}{8}
[ ] D: frac{33}{4}

Solution: Iterate: intlimits_{0}^{3}intlimits_{0}^{1}intlimits_{0}^{1} leftx y + zright, dx, dy, dz = frac{21}{4} .

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Q7. Evaluate iiint_B x y + z,dV over the box B=[0,1]times[0,3]times[0,3] .

[ ] A: frac{81}{2}
[ ] B: frac{81}{8}
[x] C: frac{81}{4}
[ ] D: frac{117}{4}

Solution: Iterate: intlimits_{0}^{3}intlimits_{0}^{3}intlimits_{0}^{1} leftx y + zright, dx, dy, dz = frac{81}{4} .

Q8. Evaluate iiint_B x + y + z,dV over the box B=[0,3]times[0,3]times[0,1] .

[ ] A: 63
[x] B: frac{63}{2}
[ ] C: frac{63}{4}
[ ] D: frac{81}{2}

Solution: Iterate: intlimits_{0}^{1}intlimits_{0}^{3}intlimits_{0}^{3} leftx + y + zright, dx, dy, dz = frac{63}{2} .